THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 8: **Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The 4th and 9th terms of an AP are 14 and 4. Its 15th term is: 1

(a) -7 **(b)** -8 **(c)** -6 **(d)** -9

2. The sum of the first 17 even natural numbers is: 1

(a) 289 **(b)** 323 **(c)** 340 **(d)** 306

3. The sum of the first 19 odd natural numbers is: 1

(a) 380 **(b)** 361 **(c)** 399 **(d)** 342

4. The ratio of the 5th term to the 3rd term of a GP is 16. The common ratio (positive) is: 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** 5

5. The number of multiples of 9 between 30 and 200 (both excluded) is: 1

(a) 20 **(b)** 19 **(c)** 21 **(d)** 18

6. The value of the 20th term minus the 10th term of the AP $8, 5, 2, -1, \ldots$ is: 1

(a) -27 **(b)** -20 **(c)** -33 **(d)** -30

7. In an AP, $t_1 = 2$ and $t_6 = 67$. The common difference is: 1

(a) 15 **(b)** 14 **(c)** 13 **(d)** 12

8. In an AP, $d = 11$ and $t_7 = 70$. The first term is: 1

(a) 5 **(b)** 4 **(c)** 3 **(d)** 6

9. The 4th and 9th terms of an AP are 21 and 46. Its 15th term is: 1

(a) 76 **(b)** 78 **(c)** 75 **(d)** 77

10. The sum of the first 22 even natural numbers is: 1

(a) 550 **(b)** 506 **(c)** 528 **(d)** 484

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the sum of all multiples of 8 between 20 and 150. 2

12. The second and fifth terms of a GP are 10 and 80. Find the common ratio and the first term. 2

13. The 6th term of an AP is 37 and the 15th term is 100. Find d. 2

14. Find the sum of all multiples of 12 between 30 and 200. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the number of terms of the AP $13, 22, 31, \ldots$ whose last term is 283, and its sum. 3

16. The 5th term of an AP is 37 and the 7th term is 55. Find the AP and its 25th term. 3

17. Find the sum of the first 12 terms of the AP whose first term is 3 and whose 6th term is 43. 3

OR

Find the number of terms of the AP $-20, -15, -10, \ldots$ whose last term is 155, and its sum. 3

18. The 4th term of an AP is 12 and the 10th term is 60. Find the AP and its 18th term. 3

OR

Find the sum of the first 16 terms of the AP whose first term is -5 and whose 9th term is 27. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A ladder has 9 rungs. The bottom rung is 100 cm long and each rung above is 5 cm shorter than the one below. Find the length of the top rung and the total length of wood needed for the rungs. 5

OR

A sum of ₹50000 is invested at 20% per year compound interest. Show that the amounts at the end of years form a GP and find the amount after 3 years. 5

20. A ladder has 13 rungs. The bottom rung is 150 cm long and each rung above is 6 cm shorter than the one below. Find the length of the top rung and the total length of wood needed for the rungs. 5

OR

A sum of ₹8000 is invested at 25% per year compound interest. Show that the amounts at the end of years form a GP and find the amount after 2 years. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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